Microeconomics Midterm
Fall 2026

Student ID:

Name:

Instructions

1. Do NOT flip over this page until every student receives this exam. Your TA will let you know when you can start.
2. During this closed-book exam, you cannot consult any materials.
3. If you are unable to explain your reasoning in English, it is okay to write in Korean.
4. There are 12 questions in total. Both sides are printed, so make sure you check the backside of each page.

Honor Code: Cheating on exams or quizzes, plagiarizing someone else's answers as one's own, or any other instance of academic dishonesty violates the standards of academic integrity.

Confidentiality Code: Sharing the information of the exam or quiz contents with other students in any form and medium is strongly prohibited, as it raises information inequity.

I, _____, consent to the Honor Code and the Confidentiality Code.
(write your name)

1. Suppose the equilibrium price of gasoline is \$4 per gallon. Suddenly, two events occur simultaneously: the price of electric vehicles (a substitute for gasoline-powered cars) decreases significantly, and a major oil refinery shuts down due to a hurricane. What happens in the market for gasoline?

1. Both equilibrium price and quantity will decrease.
2. The equilibrium price will increase, but the effect on quantity is ambiguous.
3. The equilibrium quantity will decrease, but the effect on price is ambiguous.
4. Both equilibrium price and quantity will increase.

2. A consumer spends her income on goods A and B. Determine whether each statement is **TRUE**, **FALSE**, or **UNCERTAIN**. Briefly explain your reasoning.

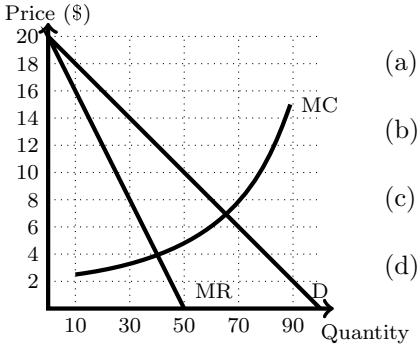
(a) If the price of good A increases, the substitution effect will always cause the consumer to buy less of good A, regardless of whether A is a normal or inferior good.

(b) If good A is an inferior good and its price increases, the income effect will prompt the consumer to buy less of good A.

(c) Good A is considered a Giffen good if the magnitude of the substitution effect outweighs the magnitude of the income effect when its price changes.

3. You own a specialized 3D printer purchased for \$9,000. You have three options for a weekend project: (A) rent it to a friend for \$400, (B) print custom figurines yielding \$500 in sales but requiring \$150 in materials, or (C) donate it to a local school for a tax write-off worth \$300. You already paid a non-refundable \$50 maintenance fee for the weekend. What is the opportunity cost of choosing option B (printing figurines)?

4. The figure below shows a monopolist's demand curve (D), marginal revenue (MR), and marginal cost (MC). Evaluate whether each of the following statements is **TRUE** or **FALSE**.



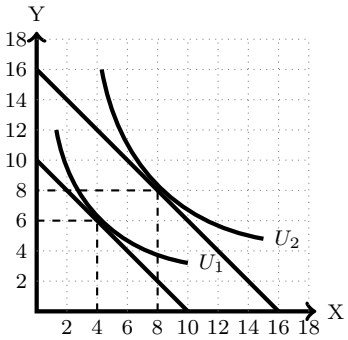
(a) If the firm produces 45 units, the marginal revenue is positive.

(b) The profit-maximizing price is \$4 where $MR = MC$.

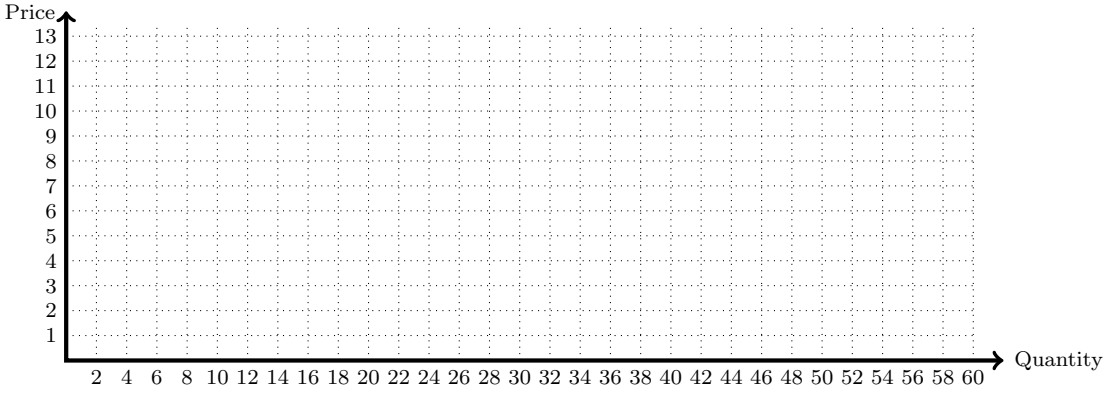
(c) At a price of \$14, the firm's profit would rise if it lowers its price.

(d) The profit-maximizing quantity is 65 units where $D = MC$.

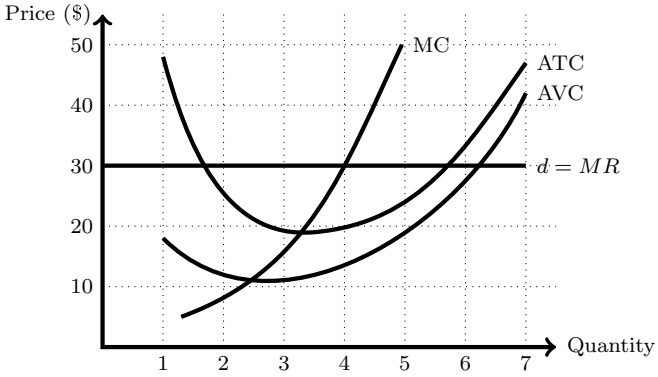
5. The income elasticity of demand for good X is $\frac{\%\Delta Q_X}{\%\Delta I}$. The figure below shows optimal consumption bundles at two different income levels, with the prices of both goods held constant at \$2 each. Based on this information provided in the figure, calculate the income elasticity of demand for good X.



6. Suppose Tom and Sarah are the only two customers in the market for guitar lessons. Tom's demand is $Q_T = 36 - 4P$, and Sarah's demand is $Q_S = 24 - 2P$. Draw the market demand curve for guitar lessons on the axes below, labeling any kink points.



7. The following figure shows a firm's short-run average total cost, average variable cost, marginal cost, and marginal revenue. At the profit-maximizing quantity, what is the firm's total cost?

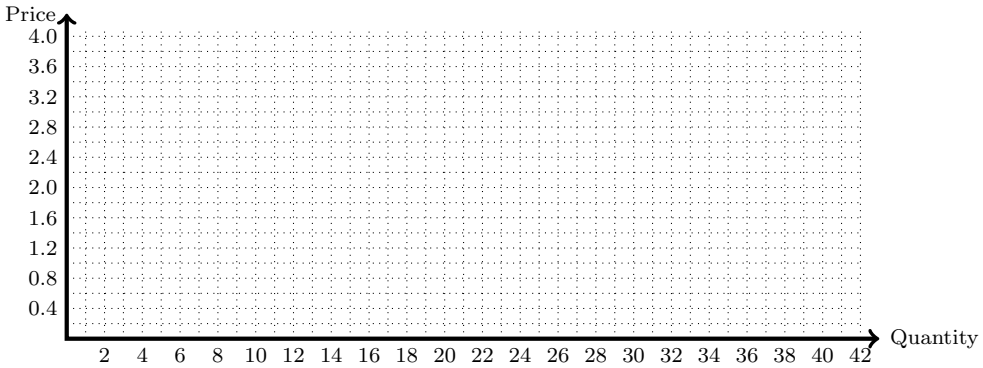


8. Suppose the market for bottled water is represented by the supply and demand functions $Q_S = 30P$ and $Q_D = 40 - 10P$, where P is the price per bottle (in dollars) and Q measures bottles per minute.

(a) Draw the supply and demand functions on a graph below. Make sure the vertical axis represents P.

(b) Find the equilibrium price and quantity. What are the values of consumer surplus (CS) and producer surplus (PS)?

(c) Suppose the government imposes a \$0.80 per-bottle tax on consumers. What is the new equilibrium quantity? What is the deadweight loss value?



9. Sofia is a monopolist who sells hand-painted ceramics that no one else can produce. Fill in the blanks in the table below.

Quantity	Price ($ per piece)	Total Revenue	Marginal Revenue
0	$600	$0	—
1	$500	_____	_____
2	_____	_____	$300
3	_____	$900	_____

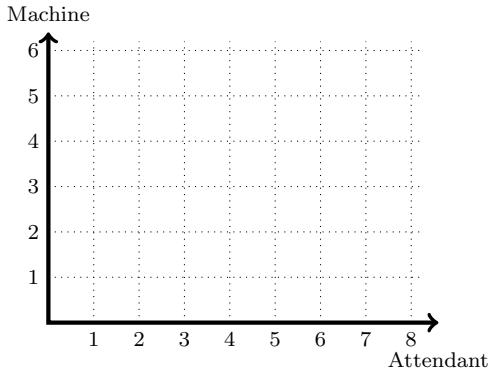
10. At a car wash, it takes 2 attendants and 1 machine to process 100 cars per hour. This ratio is fixed; you cannot substitute attendants for machines.

(a) Draw the production isoquants corresponding to 200 and 300 cars per hour in the figure to the right.

(b) Suppose each attendant earns $25 per hour and each machine rents for $50 per hour. Draw the isocost lines corresponding to total costs of $100 and $200.

(c) For 200 cars per hour, find the optimal input bundle that minimizes total operating cost.

(d) Does this production technology exhibit increasing, constant, or decreasing returns to scale? Briefly explain your reasoning.



11. The market for artisanal soap is perfectly competitive, and the equilibrium price is $20 per box. Chloe is a soap maker with a short-run fixed cost of $150 and a short-run variable cost of $2Q^2$, where Q is the number of boxes produced. Her short-run total cost is $TC = 150 + 2Q^2$ and her marginal cost is $MC = 4Q$.

(a) Chloe produces 10 boxes. What is her average variable cost and average total cost?

(b) What is Chloe's current profit at $Q = 10$?

(c) Evaluate whether Chloe should shut down or stay open. If she should stay open, suggest the optimal production level. If she should shut down, explain why.

12. Maya is a monopolist who holds a patent for a specialized solar-powered charger. She has a constant production cost of $20 for each charger she makes (both her marginal cost and average cost are $20).

(a) Suppose the market demand function for Maya's chargers is $Q = 140 - 2P$. Find the profit-maximizing quantity of chargers and the corresponding price.

(b) Suppose the market demand function for the chargers shifts to $Q = 120 - P$. Find the profit-maximizing quantity of chargers and the corresponding price.

(c) Compare your answers from (a) and (b). Explain why Maya charges different prices despite having the exact same marginal cost.